The Standard Trade Model and External Economies of Scale

ECON 308 – International Economic Relations

Fall 2025

Problem 1 — Terms of Trade and Welfare (Standard Trade Model)

Data. PPF: $F^2 + M^2 = 400$ with F on the horizontal axis and M on the vertical axis. World relative price $P_M/P_F = 2$.

(a) Value-maximizing production point

We maximize the value of output at world prices. Normalize $P_F = 1$ so $P_M = 2$ and

$$V = P_F F + P_M M = F + 2M \quad \text{s.t.} \quad F^2 + M^2 = 400.$$

Method 1 (tangency of isovalue and PPF). The slope of the PPF is the marginal rate of transformation:

$$F^2 + M^2 = 400 \Rightarrow 2F\,dF + 2M\,dM = 0 \Rightarrow \frac{dM}{dF} = -\frac{F}{M}.$$

The slope of the isovalue line is $-\frac{P_F}{P_M} = -\frac{1}{2}$. Tangency requires

$$-\frac{F}{M} = -\frac{1}{2} \quad \Rightarrow \quad \frac{F}{M} = \frac{1}{2} \quad \Rightarrow \quad F = \frac{M}{2}.$$

Impose on the PPF:

$$\left(\frac{M}{2}\right)^2 + M^2 = 400 \;\Rightarrow\; \frac{M^2}{4} + M^2 = \frac{5}{4}M^2 = 400 \;\Rightarrow\; M^2 = 320 \;\Rightarrow\; M = \sqrt{320} = 8\sqrt{5}.$$

Hence $F = \frac{M}{2} = 4\sqrt{5}$.

$$\boxed{(F_P, M_P) = \left(4\sqrt{5},\ 8\sqrt{5}\right) \approx (8.944,\ 17.889).}$$

Check (Method 2, Lagrangian).

$$\mathcal{L} = F + 2M + \lambda(400 - F^2 - M^2).$$

FOCs: $1 - 2\lambda F = 0 \Rightarrow F = \dfrac{1}{2\lambda}$; $\;2 - 2\lambda \cdot 2M = 0 \Rightarrow M = \dfrac{1}{2\lambda}$. Thus $F = M$ from the FOCs *alone*; but this contradicts the tangency condition above. The mistake is that the derivative for M is $2 - 2\lambda(2M)$ *only if* the constraint is $F^2 + (2)M^2 = \cdots$. Our actual constraint is $F^2 + M^2$, so

$$\frac{\partial \mathcal{L}}{\partial M} = 2 - 2\lambda M = 0 \Rightarrow M = \frac{1}{\lambda}, \quad \frac{\partial \mathcal{L}}{\partial F} = 1 - 2\lambda F = 0 \Rightarrow F = \frac{1}{2\lambda},$$

hence $F = \dfrac{M}{2}$, agreeing with the tangency result.

(b) Consumption with preferences $M = 0.5F$

Income (value of production) at world prices:

$$V = F_P + 2M_P = 4\sqrt{5} + 2(8\sqrt{5}) = 20\sqrt{5} \approx 44.721.$$

Budget (trade) line at world prices:

$$F + 2M = V = 20\sqrt{5}.$$

Preferences require $M = 0.5F$. Substitute into the budget:

$$F + 2(0.5F) = 2F = 20\sqrt{5} \Rightarrow F_C = 10\sqrt{5}, \quad M_C = 0.5F_C = 5\sqrt{5}.$$

$$\boxed{(F_C, M_C) = \left(10\sqrt{5},\ 5\sqrt{5}\right) \approx (22.361,\ 11.180).}$$

(c) Trade pattern and volumes

$$\text{Exports of } M:\ X_M = M_P - M_C = 8\sqrt{5} - 5\sqrt{5} = 3\sqrt{5} \approx 6.708.$$
$$\text{Imports of } F:\ M_F = F_C - F_P = 10\sqrt{5} - 4\sqrt{5} = 6\sqrt{5} \approx 13.416.$$

Value balance check at world prices:

$$\text{Export revenue } = P_M X_M = 2{\cdot}3\sqrt{5} = 6\sqrt{5}, \qquad \text{Import expenditure } = P_F M_F = 1{\cdot}6\sqrt{5} = 6\sqrt{5}.$$

Balanced trade holds (up to rounding).

$$\boxed{\text{Country exports manufactures and imports food.}}$$

(d) Effect of an increase in P_M/P_F on welfare

A rise in P_M/P_F (steeper world price line) has two effects in the Standard Trade Model:

- **Production effect:** Firms re-optimize toward the now more valuable good (M), moving to a tangency where the PPF slope equals $-P_F/P_M$ (more negative). Output bundle becomes more M-intensive.
- **Welfare effect:** The trade (budget) line through the new production point pivots outward (higher intercepts), allowing a higher indifference curve. Since the country is an exporter of M, this is an *improvement in its terms of trade* and thus raises welfare.

$$\boxed{\text{Welfare rises when } P_M/P_F \text{ increases for an } M\text{-exporter (no distortions).}}$$

Problem 2 — Export- and Import-Biased Growth

Setup. Country A exports steel (S) and imports textiles (T). Initially, $P_S/P_T = 1$. Consider biased growth holding the other country unchanged.

(a) Export-biased growth in A

Mechanics (diagrammatically):

i) *PPF / RS shift:* Export-biased growth expands A's capacity in the export sector (S), rotating the PPF outward more in the S direction. Relative supply $RS = Q_S/Q_T$ shifts *right*.

ii) *World price:* For a *large* A, the outward shift in world RS tends to *lower* P_S/P_T (the price of S relative to T).

iii) *Production & consumption:* At given prices (small-country case), A produces more S and trades for more T; consumption moves to a higher indifference curve. For a large-country, the deterioration in P_S/P_T dampens the gain.

Terms of trade (TOT) and welfare:

$$\boxed{\text{Export-biased growth tends to } \textit{worsen} \text{ Home's TOT } (P_S/P_T \downarrow).}$$

- **Small-country:** Prices fixed $\Rightarrow$ unambiguous welfare gain (bigger budget set).
- **Large-country:** Price falls for S reduce the gain via a worse TOT. Welfare typically still rises unless the TOT deterioration is extreme ("immiserizing growth").

(b) Import-biased growth in A

Mechanics (diagrammatically):

i) *PPF / RS shift:* Import-biased growth expands capacity or reduces demand for the import good T, effectively raising the world relative supply of T and/or lowering A's import demand.

ii) *World price:* For a *large* A, this tends to *raise* P_S/P_T (since T becomes relatively more abundant/cheaper internationally).

iii) *Production & consumption:* At given prices (small-country), A can reach a higher indifference curve due to a larger PPF. For a large-country, the *improved* TOT magnifies the welfare gain.

Terms of trade (TOT) and welfare:

$$\boxed{\text{Import-biased growth tends to } \textit{improve} \text{ Home's TOT } (P_S/P_T \uparrow).}$$

- **Small-country:** Prices fixed $\Rightarrow$ welfare gain from the outward shift alone.
- **Large-country:** Additional welfare gain from better TOT (steeper trade line).

(c) Summary

$$\boxed{\text{Export-biased growth: welfare} \uparrow \text{but TOT} \downarrow \text{(for a large exporter).}}$$

$$\boxed{\text{Import-biased growth: welfare} \uparrow \text{and TOT} \uparrow \text{(for a large exporter).}}$$

In both cases, small countries (price takers) experience unambiguous welfare gains because world prices are unaffected by their growth.